

A PROJECT REPORT
ON
CADesk
SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE DEGREE
BACHELOR OF COMPUTER APPLICATION (BCA)

SUBMITTED BY
SANCHITA BAROKAR
ARCHITA KURADE
ARPITA KURADE
TANVI KHANDARE
PURVA SHINDE

UNDER THE GUIDANCE OF
MRS. MANASI GHULE
MS. VAIDEHI RAVAL
MS. SAKSHI PATIL



JANSEVA SAMITI SANCHALIT
SHRI. M. D. SHAH MAHILA COLLEGE OF ARTS AND COMMERCE (Autonomous)
(AFFILIATED COLLEGE TO S.N.D.T WOMEN'S UNIVERSITY) B.J. PATEL ROAD
MALAD (WEST)
MUMBAI-400064
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DECLARATION

We Ms. Sanchita Barokar, Ms. Archita Kurade, Ms. Arpita Kurade, Ms. Tanvi Khandare, Ms. Purva Shinde students of Bachelor of Computer Applications (BCA) Semester VI (2025-26), hereby declare that the Project Report submitted to Shri M. D. Shah Mahila College has been entirely developed by us to fulfil the requirements of the final year project.

We affirm that this project has been completed within the specified time frame. The information provided by us is true and original to the best of our knowledge.

Sanchita Sanjay Barokar - _____

Archita Vitthal Kurade - _____

Arpita Vitthal Kurade - _____

Tanvi Santosh Khandare - _____

Purva Rajaram Shinde - _____

ACKNOWLEDGEMENT

We started this project as a part of the curriculum of B.C.A. which was assigned to us by "SHRI M.D. SHAH MAHILA COLLEGE". It was very exciting for us to work on our project "CADesk".

We would like to express our sincere gratitude to everyone who contributed to the success of our project.

First, we extend our heartfelt thanks to SHRI M.D. SHAH MAHILA COLLEGE for providing us with the necessary resources and a conducive learning environment, which greatly aided the development of our project.

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We are also thankful to Principal Dr. Deepa Sharma and the College Management for their ongoing support and the facilities provided to us.

Finally, we would like to thank our friends for their positive criticism and encouragement, which kept us motivated and helped us refine our work.

Thank you to all who played a part in this project—your contributions have been truly appreciated.

College Certificate

Client Certificate

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1. SYNOPSIS

(1) About Us

1.1 Project Name:

CA Client Documents Access Web Application

Application Name: CADesk

1.2 Member details:

- *Sanchita Sanjay Barokar (66)*
- *Archita Vitthal Kurade (26)*
- *Arpita Vitthal Kurade (27)*
- *Tanvi Santosh Khandare (24)*
- *Purva Rajaram Shinde (46)*

Project Introduction

In today's digital world, Chartered Accountants (CA) and their clients deal with many important financial documents such as Income Tax Return (ITR) files, Goods and Services Tax (GST) files, and Tax Deducted at Source (TDS) records. Traditionally, these documents are shared using physical copies, emails, or messaging applications such as WhatsApp. This process is time-consuming, insecure, and difficult to manage.

The CA Client Documents Access Website is designed to solve these problems by providing a secure, centralized, and easy-to-use online platform. The application allows users to log in securely and access their documents under predefined categories such as ITR, GST, and TDS. Users can view and download the required files anytime and anywhere.

This project aims to reduce paperwork, improve document security, and enhance communication between Chartered Accountants and their clients.

(2) Client Details (Overview)

- **Client Name:** Purushottam Gohil
- **Client Type:** Chartered Accountant Firms
- **Target Users:**
 - Chartered Accountants
 - CA Office Staff
 - Registered Clients

- **Application Type:** Website
- **Supported Platform:** Any Web Brower
- **Purpose:** Secure access and download of financial documents

2.1 Client Details:

The primary clients of this project are **Chartered Accountants** who manage multiple clients and large volumes of tax-related documents. The secondary users are the **clients of the CA**, who require quick and secure access to their documents.

Problems Faced in the Existing System

- Difficulty in accessing documents on time
- Dependency on emails and physical copies
- Risk of data loss or unauthorized access
- No centralized document management system
- Time-consuming document retrieval

Proposed Solution

- Secure login-based Website
- Centralized document storage
- Categorization of documents (ITR, GST, TDS)
- Easy and fast file download facility
- Admin-based client management system

2.2 Objective and Purpose of the Project

Objectives

- To develop a secure website for CA document access
- To provide categorized access to ITR, GST, and TDS files

- To reduce manual paperwork
- To allow users to download required documents easily
- To enable admin to add and manage clients

Purpose

- Improve efficiency in CA offices
- Ensure data confidentiality and security
- Save time for both CA and clients
- Provide a paperless and digital solution

2.3 Comparison: Existing System vs Proposed System

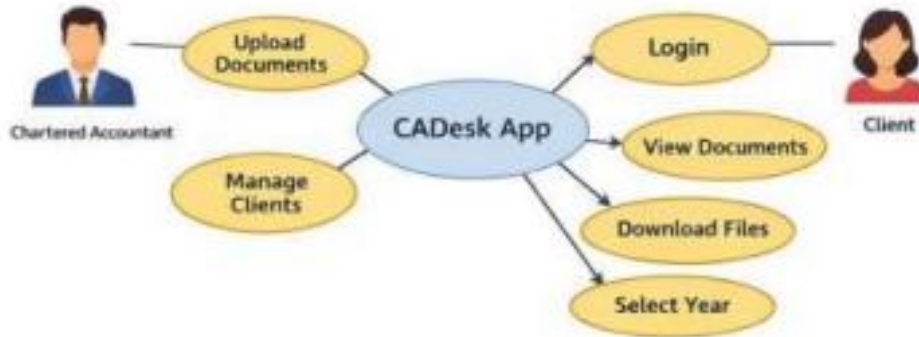
Existing System

- Manual paperwork
- Documents shared via email or WhatsApp
- High risk of data loss
- Low security
- Time-consuming process

Proposed System

- Fully digital document management
- Secure login-based access
- Cloud-based storage
- Categorized documents (ITR, GST, TDS)
- Instant file download

2.4 Scope and Features (Module Details)



Scope of the Project

- Small and medium Chartered Accountant firms
- Individual CA practitioners
- Secure access to tax-related documents such as ITR, GST, and TDS
- Easy document download facility with proper authentication

(3) MODULES

Module 1: Landing Page Module

This is the first screen of the Website

Functions:

- Displays application name “CADesk”
- Attractive and colourful user interface
- Shows animations, and welcome message
- Contains “Get Started” button
- Redirects user to Role Selection page

Purpose:

To introduce the application and provide an entry point for users.

Module 2: Role Selection Module

This module allows the user to choose their role in the system.

Features:

- Displays two options: Admin/Client
- User selects appropriate role
- Navigates to respective login page

Purpose:

To separate admin and client access and route users correctly.

Module 3: Login / Authentication Module

This module handles authentication and security.

Functions:

- Separate login for Admin and Client
- User ID / Email and Password fields
- Input validation
- Firebase Authentication integration
- Error alerts such as :“Incorrect username or password
- Logout functionalit

Purpose:

To ensure that only authorized users can access the system securely.

Module 4: Admin Module

This module is used only by the Admin (CA office).

Admin Features:

- Admin Dashboard

- Upload documents to Firebase Storage
- Assign documents to specific clients
- View list of all registered clients
- View uploaded documents of any client
- Add new client directly from dashboard
- Register client details (Name, Email)
- Generate login credentials
- Manage client list

Purpose:

To allow the CA/Admin to manage client documents efficiently.

Module 5: Client Module

This module is for client-side operations.

Client Features:

- Secure login using registered credentials.
- Client dashboard with 3 categories: GST, ITR, TDS
- Year selection (e.g., Last 1 year , Last 2 year and Both)
- View only their own documents
- Download document
- No permission to upload files

Purpose:

To provide clients secure access to their personal tax-related documents.

Module 6: Profile & Logout Module

This module manages user profile and session control.

Features:

- Display client profile details:
 - Name
 - Email
- Modern profile layout using cards
- Secure logout option
- Confirmation dialog before logout

Purpose:

To manage user information and provide safe exit.

3.1 Front-End Technology

- HTML
- CSS
- JavaScript

3.2 Back-End Technology

- Firebase Authentication
- Firebase Fire store
- Firebase Cloud Storage

3.3 Database

- Firebase Firestore (Cloud Database)

(4) System Requirements

System requirements specify the hardware and software resources needed to develop, install, and run the CA Client Documents Access website efficiently.

They ensure that the system performs smoothly without errors or performance issues.

4.1 Hardware Requirements

a) Laptop / Desktop Computer

- A computer system is required to develop and access the website.
- The system should support modern web browsers and development tools.
- Used by CA, office staff, and clients to access documents through the browser.

b) RAM

Minimum required RAM: 4 GB (Recommended).

Sufficient RAM ensures:

- Smooth performance
- Faster loading of web pages
- Efficient running of development tools
- No system hanging while uploading or downloading files

c) Storage Space

Adequate internal storage is required to:

- Store project files
- Install development tools
- Download client documents

At least 500 MB free space is recommended for proper working.

d) Internet Connectivity

A stable internet connection is required.

Used for:

- User authentication
- Accessing Firebase database
- Uploading and downloading documents
- Real-time data synchronization

Supports Wi-Fi or mobile hotspot connection.

4.2 Software Requirements

a) Operating System

Any modern operating system can be used to run the web application.

Supported Operating Systems:

- Windows 10
- Windows 11
- macOS
- Linux

The operating system must support modern web browsers and development tools.

b) Web Browser

A web browser is required to access and test the website.

Supported in all Browser

Browsers are used for:

- Running the CADesk website
- Testing user interface
- Downloading documents

c) Code Editor

A code editor is required for developing the web application.

Recommended Tool:

VS Code (Visual Studio Code)

Used for:

- Writing HTML, CSS, and JavaScript code
- Editing project files
- Debugging and testing

d) Front-End Technologies

The following technologies are used for building the user interface:

- **HTML** – for creating web page structure
- **CSS** – for designing and styling pages

- **JavaScript** – for logic and interactivity

These technologies help to create an interactive and user-friendly web application.

e) Firebase Services

Firebase is used as Backend-as-a-Service (BaaS) for the project.

Firebase Services Used:

- **Firebase Authentication** – for secure login
- **Firebase Firestore** – for storing client and document data
- **Firebase Cloud Storage** – for storing uploaded files

Firebase helps in:

- Secure data management
- Real-time database operations
- Cloud-based storage
- Easy backend integration

f) Internet Browser (For Development)

Required for:

- Accessing Firebase console
- Managing database and authentication
- Testing the application in real-time

2. Software Development Life Cycle (SDLC)

The Software Development Life Cycle (SDLC) is a structured framework used by software organizations to design, develop, and test high-quality software. It defines the entire process of software production, from the initial idea to the final deployment and maintenance.

The goal of the SDLC is to produce high-quality software that meets or exceeds customer expectations, reaches completion within time and cost estimates, and is efficient to maintain.



Why is SDLC Important?

Without a structured SDLC, software development becomes chaotic. Teams may miss requirements, blow budgets, or deliver buggy code that doesn't solve the user's problem. SDLC provides:

- **Visibility:** Stakeholders know exactly what is happening at every stage.
- **Quality Control:** Testing is integrated into the process, not an afterthought.
- **Risk Management:** Potential pitfalls are identified during the planning phase.
- **Cost Estimation:** Helps in predicting timelines and budgets accurately.

Stages of the Software Development Life Cycle

SDLC specifies the tasks to be performed at various stages by a software engineer or developer. It ensures that the end product is able to meet the customer's expectations and fits within the overall budget. Hence, it's vital for a software developer to have prior knowledge of this software development process. SDLC is a collection of these six stages, and the stages of SDLC are as follows:



Software Development Life Cycle Model SDLC Stages

The SDLC Model involves six phases or stages while developing any software.

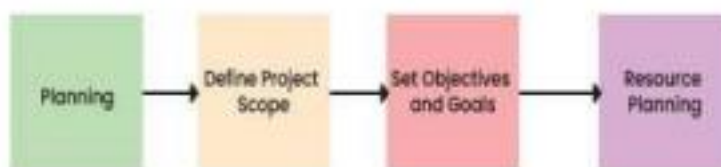
Stage 1: Planning and Requirement Analysis

This is the most fundamental stage.

Before writing code, the team must define what they are building and why.

- **Activities:** Feasibility studies (technical, operational, economic), resource allocation, project scheduling, and cost estimation.
- **Output:** Project Plan, Feasibility Report.
- **Key Players:** Senior Engineers, Project Managers, Stakeholders.

Stage-1: Planning and Requirement Analysis



6 Stages of Software Development Life Cycle

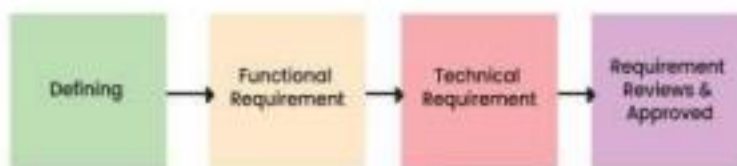
Stage 2: Defining Requirements (SRS)

Once the plan is approved, the specific product requirements must be defined and documented. This is done through the **Software Requirement Specification (SRS)** document.

- **Activities:** Gathering detailed functional and non-functional requirements from customers.
- **Output:** SRS Document (The "Bible" for the development team).

Key Players: Business Analysts (BAs), Product Owners

Stage-2: Defining Requirements



6 Stages of Software Development Life Cycle



Stage 3: Designing Architecture

The requirements are turned into a technical blueprint. This phase defines the overall system architecture and technology stack.

- **High-Level Design (HLD):** Defines the architecture, database design, and relationships between modules.
- **Low-Level Design (LLD):** Defines the logic of individual components, API interfaces, and database tables.
- **Output:** Design Document Specification (DDS).
- **Key Players:** System Architects, Lead Developers.

Stage-3: Designing Architecture



6 Stages of Software Development Life Cycle

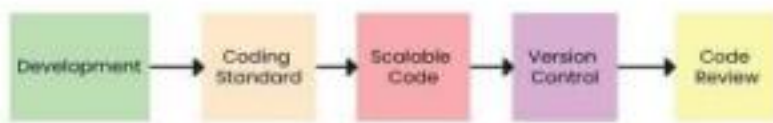


Stage 4: Development (Coding)

This is the longest phase where the actual building takes place. Developers write code based on the Design Document.

- **Activities:** Coding, Code Reviews, Unit Testing, and Static Code Analysis.
- **Tools:** Compilers, Debuggers, IDEs (VS Code, IntelliJ), Version Control (Git).
- **Output:** Source Code, Executable Software.
- **Key Players:** Developers (Frontend, Backend, Full Stack).

Stage-4: Developing Product



6 Stages of Software Development Life Cycle

Stage 5: Testing

Once the code is written, it moves to the Quality Assurance (QA) team. The goal is to find bugs before the customer does.

Types of Testing:

- **Unit Testing:** Testing individual functions.
- **Integration Testing:** Ensuring modules work together.
- **System Testing:** Testing the entire application flow.
- **User Acceptance Testing (UAT):** Verifying the software meets business needs.

Output: Bug Reports, Test Cases, Quality Report.

Key Players: QA Engineers, Testers.

Stage-5: Product Testing and Integration



6 Stages of Software Development Life Cycle

Stage 6: Deployment & Maintenance

The software is deployed to the live environment and maintained regularly.

- **Activities:** Deploying application to production, Smoke testing , Bug fixing and updates , Performance improvement
- **Output:** *Live application.*
- **Key Players:** *DevOps Engineers, Developers, Support Team*



3. System Planning

A system is a collection of interrelated components that work together to achieve a common goal.

System planning is the initial stage of the project where the overall plan for developing the system is prepared.

In this phase, we decide what to build, how to build, and how to manage time and resources for the project.

What we do in system planning

1. Define Project Scope

Decide features like:

- Admin login
- Client login
- Upload and download of ITR, GST, TDS documents
- Secure document storage
- Add New Client option
- Web-based access through browser

2. Identify Objectives

- Provide secure access to client documents
- Reduce manual paperwork
- Improve data safety and accessibility
- Allow clients to access documents anytime
- Enable admin to manage clients easily

3. Resource Planning

Hardware:

- Laptop / Desktop
- Internet connection

Software:

- VS Code
- Web Browser
- Firebase services

Human Resources:

- Project team members
- Guide and faculty support

4. Time Planning (Scheduling)

- Prepare project timetable using Gantt Chart
- Allocate sufficient time for:
 - Frontend development
 - Firebase integration
 - Testing
 - Documentation

5. Feasibility Consideration

- Technical feasibility (HTML, CSS, JS + Firebase support)
- Economic feasibility (free tools and cloud services)
- Operational feasibility (easy for CA and clients to use)

6. Risk Identification

- Data security issues
- Delay in coding
- Internet dependency
- Browser compatibility problems

3.1 Gantt Chart



1. Project Search -

Research is done to understand similar systems, technologies, and possible solutions before finalizing the project approach.

2. Requirement Analysis -

All functional and non-functional requirements are identified, including user needs, features, and constraints.

3. System Planning -

System planning is the process of deciding the project's scope, objectives, resources, schedule, and feasibility before starting the actual development of the system.

4. System Analysis -

The proposed system is analyzed to understand workflows, data flow, and system behavior.

5. System Design -

The overall system architecture is designed, including modules, database structure, and interfaces.

6. System Coding -

The Actual development of the system begins, where code is written based on the design specifications.

7. System Testing -

The system is tested to identify and fix errors, ensuring it works as expected.

8. Implementation -

The tested system is deployed in the real environment and made available to users.

9.Submission -

Final project submission is done, including source code, documentation, reports, and presentation.

4. System Analysis

Introduction:

System Analysis is the process of studying the existing manual system, understanding its problems, and finding the requirements for developing a computerized system.

In the CA Desk project, system analysis helps us understand what CAs and their clients need for secure and efficient document management.

The main purpose of system analysis is to clearly understand user needs before starting the actual development of the application.

4.1 Requirement Gathering

Requirement Gathering is the process of collecting information about what the system should do and how it should work. It helps the developer understand the needs of users before starting the development.

In the CA Client Documents Access web Application, requirements were gathered by understanding how Chartered Accountants and their clients currently manage and access documents.

How requirements were collected:

- By discussing with CA office staff
- By observing manual document handling
- By studying problems faced in the existing system
- By understanding client requirements

Functional Requirements

System must provide separate login for Admin and Client

System must authenticate users using Firebase Authentication

Admin must be able to add new clients

Admin must be able to upload documents

Admin must be able to assign documents to clients

Admin must be able to view and manage client records

Clients must be able to log in securely

Clients must be able to view their own documents only

Clients must be able to download documents

Documents must be categorized as ITR, GST, and TDS

Year-wise document selection must be available

System must provide logout option

Non-Functional Requirements

System must be secure and protect user data

Application should load quickly

User interface must be simple and easy to use

System should be reliable and error-free

Application must work on all modern web browsers

System should be available anytime with internet

Data should be stored safely in Firebase

4.2 Requirement Model (Software Development Model)

A Requirement Model explains how the software will be developed step by step.

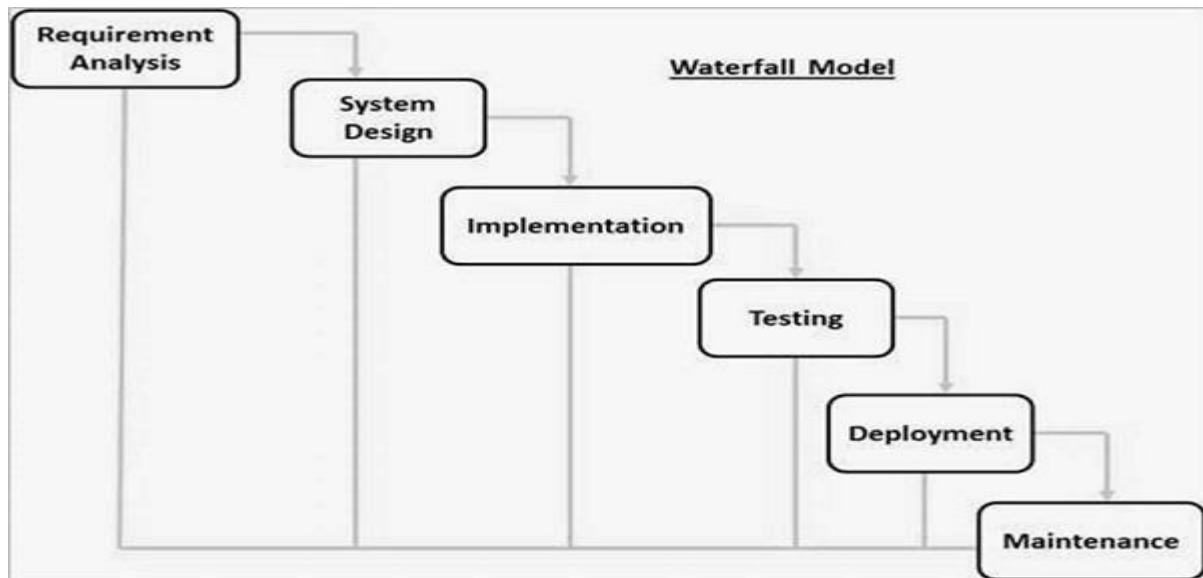
For this project, the **Waterfall Model** is used.

The **Waterfall Model** is chosen because the project requirements are clearly defined and do not change frequently.

Phases of Waterfall Model

1. Requirement Analysis – understanding user needs
2. System Design – planning system structure
3. Coding – developing the website and backend
4. Testing – checking for errors
5. Implementation – installing and running the system
6. Maintenance – fixing issues and updates

Each phase is completed before moving to the next phase.



4.3 Feasibility Study

Feasibility Study is done to check whether the project can actually work in real life and whether it is practical to develop and use.

For our CADesk Website , feasibility study is divided into three types: Functional, Technical, and System Feasibility.

a) Functional Feasibility

Functional feasibility checks if the system performs all required functions.

In our project:

- Users can log in securely
- Clients can view and download documents
- Admin can manage clients and files
- System meets all basic needs of CA and clients

Advantages

- Easy and fast document access
- Reduces manual work
- Improves efficiency

Limitations

- System works only with internet connection

b) Technical Feasibility

Technical feasibility checks whether the required technology and tools are available.

Technologies used:

- HTML, CSS, JavaScript
- Firebase Authentication
- Firebase Firestore
- Firebase Cloud Storage

Advantages

- All tools are easily available
- No need for separate server setup
- Easy to develop and maintain

Limitations

- Internet is compulsory
- Basic technical knowledge required

c) System Feasibility

Operational feasibility checks whether the system is easy to use in real life.

In this project:

- CA staff can manage clients easily
- Clients can access documents anytime
- System is simple and user friendly

Advantages

- Saves time for CA and clients
- Paperless system
- Easy to operate

Limitations

- Users must know basic computer usage

4.4 Tools and Technology

Tools and technologies are the main components used to design, develop, test, and run the CADesk Website.

This section explains all the technologies and tools used in the project in detail.

A) TECHNOLOGIES

Technologies are the core building blocks of the system that are used to develop and run the website.

1. HTML (Hyper Text Markup Language)

HTML is used to create the basic structure of the web application.

Purpose in the Project:

- Used to design web pages
- Creates structure of forms, buttons, tables, and layouts
- Displays content such as text, images, and links

Advantages:

- Easy to learn and use
- Supported by all web browsers
- Provides clear structure to web pages

Limitations:

- Cannot perform logical operations
- Used only for structure, not for styling or programming

2. CSS (Cascading Style Sheets)

CSS is used to design and style the web application.

Purpose in the Project:

- To make the user interface attractive
- To apply colors, fonts, layouts, and animations
- To design responsive web pages

Advantages:

- Improves look and feel of the website
- Helps to create professional UI
- Makes pages responsive for different devices

Limitations:

- Cannot perform programming logic
- Complex designs may require advanced CSS

3. JavaScript

JavaScript is used to add functionality and logic to the website.

Purpose in the Project:

- Handles user interactions
- Validates login forms
- Connects frontend with Firebase backend
- Controls dynamic operations like file upload and download

Advantages:

- Makes web pages interactive
- Runs directly in the browser
- Easy to integrate with Firebase

Limitations:

- Code may become complex for large website
- Depends on browser compatibility

4. Firebase (Backend Technology)

Firebase is used as a complete backend service for the website. It works as Backend-as-a-Service (BaaS) and removes the need for a separate server.

Firebase Services Used in the Project:

a) Firebase Authentication

Used for secure login system.

Functions:

- Admin and client login
- Email and password authentication
- User session management

Advantages:

- Highly secure
- Easy to implement
- No need to build custom authentication system

b) Firebase Firestore (Database)

Firestore is used as a cloud database.

Purpose in the Project:

- Stores client details
- Stores document information
- Manages categories and years
- Provides real-time data access

Advantages:

- Cloud-based storage
- Fast data retrieval
- No need to manage SQL server

Limitations:

- Requires internet connection

c) Firebase Cloud Storage

Used to store uploaded client documents.

Purpose:

- Securely store PDF and other files
- Provide download links to clients
- Manage large files easily

Advantages:

- Safe and reliable
- Easy file management
- Direct integration with Firebase

B) TOOLS

Tools are the software applications used to develop, test, and run the system.

1. VS Code (Visual Studio Code)

VS Code is the main code editor used for development.

Purpose in the Project:

- Writing HTML, CSS, JavaScript code
- Managing project files
- Debugging errors
- Running the application locally

Advantages:

- Lightweight and fast
- Free and open-source
- Supports extensions
- Easy to use

Limitations:

- Requires plugins for advanced features

2. Web Browser

Web browsers are used to run and test the application.

Browsers Used:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox

Purpose:

- Running the CADesk website
- Testing UI and functionality
- Checking responsiveness

Advantages:

- No installation required for users
- Works on any device

3. Firebase Console

Firebase Console is an online tool used to manage backend services.

Purpose:

- Managing authentication
- Managing Firestore database
- Managing cloud storage
- Monitoring users and data

Advantages:

- Easy backend management
- Real-time database control
- Secure environment

4.5 Software Specification

Software specification is a detailed description of the software requirements needed to develop and run a system.

It explains what the software should do, how it should behave, and the environment in which it will operate.

Software specification acts as a blueprint for software development and helps developers understand the exact needs of the system before implementation.

1. Operating System

The operating system provides the platform to develop and run the website.

The CAdesk Website can be developed and executed on the following operating systems:

- Windows 10
- Windows 11
- macOS
- Linux

Any operating system that supports modern web browsers and development tools can be used.

2. Programming Languages / Technologies Used

1) HTML

HTML is used for creating the structure of web pages.

It is responsible for:

- Designing forms
- Creating buttons and tables
- Displaying text and images
- Structuring the web application layout

2) CSS

CSS is used for designing and styling the web pages.

It is used to:

- Apply colors and fonts
- Design layouts
- Make the application responsive
- Improve the look and feel of the system

3) JavaScript

JavaScript is used as the main programming language for logic.

It is responsible for:

- Handling user actions
- Validating input forms
- Connecting frontend with Firebase
- Managing dynamic operations such as file upload and download

JavaScript makes the web application interactive and user-friendly.

4) Firebase

Firebase is used as the backend platform for the project.

It provides the following services:

- Firebase Authentication – for secure login
- Firebase Firestore – for storing data
- Firebase Cloud Storage – for storing documents

Firebase helps in managing user data and files without maintaining a separate server.

3. Database

Firebase Firestore Database

Firestore is used as a cloud database for the application.

It stores:

- Admin details
- Client details
- Document records
- Category information
- Year information

Firestore provides fast and real-time data access for the system.

4. Development Tools

1) VS Code

Visual Studio Code is used as the main development environment.

It is used for:

- Writing HTML, CSS, and JavaScript code
- Managing project files
- Debugging errors
- Running the application

2) Web Browser

Web browsers such as:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox

are used to:

- Run the application
- Test user interface
- Check functionalit

3) Firebase Console

Firebase Console is used to manage backend services such as:

- Authentication
- Database
- Cloud Storage
- User management

5. Server / Cloud Platfrom

The CADesk system does not require a separate server.

Firebase is used as a cloud backend service to handle:

- Data storage
- User authentication
- File management

This makes the system easy to deploy and maintain.

5. System Design

System design explains how the system will be built and how different parts will work together.

System design is the stage in which we plan and describe how the entire system will be created and how it will work. In this phase, the requirements collected during system analysis are converted into a clear and detailed plan.

System design decides:

- Structure of the system
- Different modules
- Data flow
- User interface
- Database design
- Communication between system components

For the CADesk Website , system design helps in creating a proper architecture for managing client documents securely.

5.1 Decomposition Diagram



The decomposition diagram of the CADesk Website represents the breakdown of the system into its major functional components.

At the highest level, the system is represented by the CADesk Web Application, which is decomposed into the following main components:

1. Landing Page

Displays website name and welcome message
Provides “Get Started” option
Redirects users to role selection page

2. Role Selection

Allows user to choose role: Admin or Client Directs user to appropriate login page

3. Login / Authentication

Handles login for Admin and Client
Uses Firebase Authentication
Validates email and password
Manages user session Provides logout option

4. Admin Module

- Admin Dashboard
- Upload documents
- Assign documents to clients
- View client list
- Manage documents
- Add New Client feature

This module gives full control to the CA to manage client data and files.

5. Client Module

- Client Dashboard
- View assigned documents
- Download files
- Select documents by category and year

This module allows clients to securely access their own records.

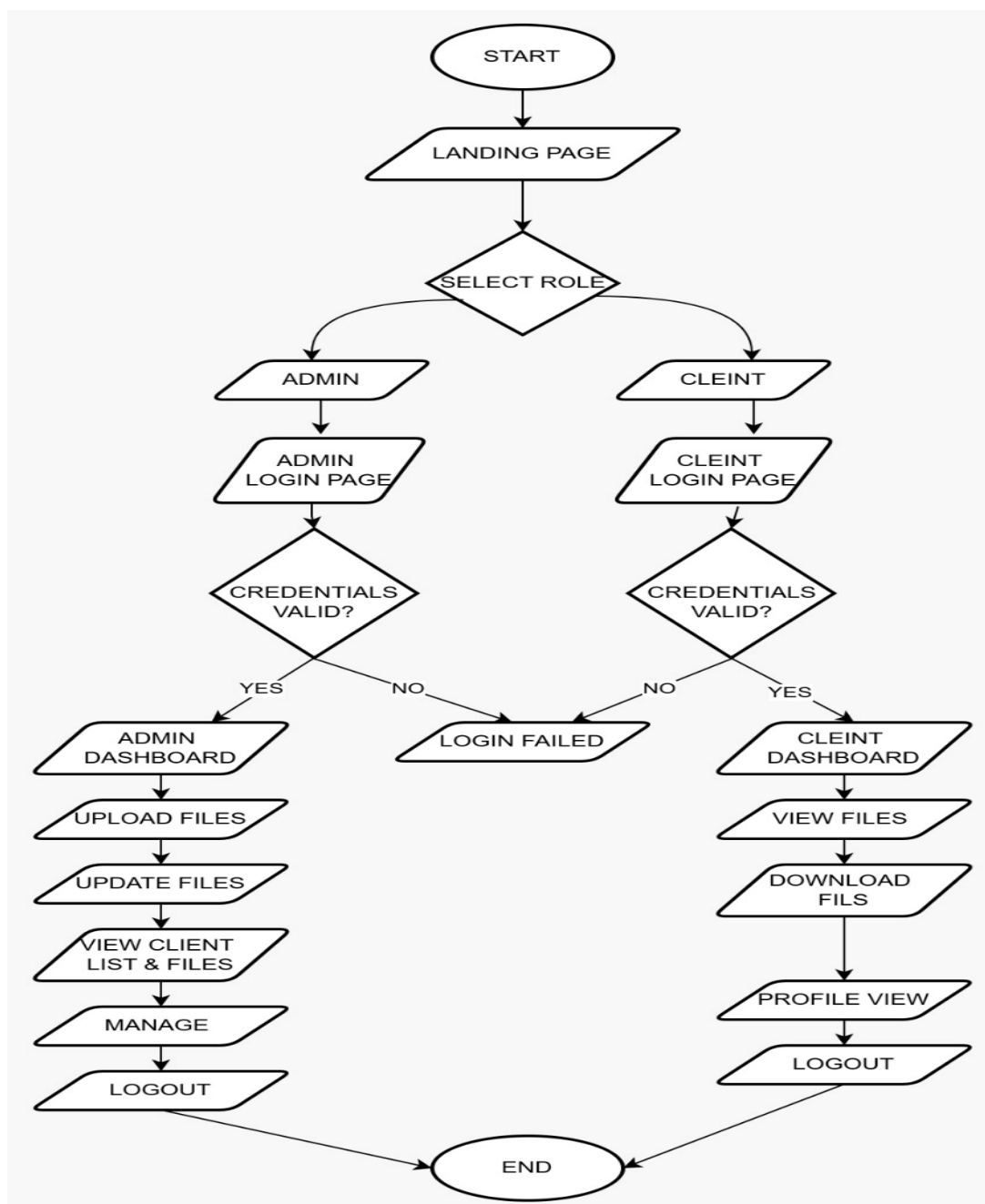
6. Profile & Logout

View user profile
Secure logout functionality

5.2 Flowchart

A **flowchart** is a diagram that shows the step-by-step working process of a system using symbols and arrows.

It explains **how a program or system works from start to end** in a simple visual form.



Purpose of a Flowchart

Flowcharts are used to:

- Understand system logic easily
- Show the flow of control in a program
- Represent decision-making steps
- Explain working of a system clearly
- Help in planning and designing software

Flowchart of CADesk Website

The flowchart of the CADesk Website represents the complete working process of the system from start to end.

1.1.1 Step-by-Step Flowchart Explanation

1. **Start**
2. User opens the CADesk Website.
3. **Landing Page** is displayed.
4. User clicks on “**Get Started**” button.
5. System navigates to **Role Selection Page**.
6. User selects role:
 - Admin
 - Client

1.1.2 Decision 1: Role Selection

7. If user selects **Admin**
 - Go to Admin Login Page
8. If user selects **Client**
 - Go to Client Login Page

1.2 ADMIN FLOW

9. Admin enters email and password.
10. System checks login credentials.

1.2.1 Decision 2: Admin Login Validation

11. If credentials are **Invalid**
 - Show “Login Failed” message
 - Return to Admin Login Page
12. If credentials are **Valid**
 - Open **Admin Dashboard**

1.2.2 Admin Dashboard Operations

13. Admin can perform the following tasks:

- Add New Client
- Upload Documents
- Assign Documents to Clients
- View Client List
- Manage Documents

14. After completing work, Admin clicks **Logout**.

15. System ends.

1.3 CLIENT FLOW

16. Client enters login credentials.

17. System checks client login details.

1.3.1 Decision 3: Client Login Validation

18. If credentials are **Invalid**

- Show “Login Failed” message
- Return to Client Login Page

19. If credentials are **Valid**

- Open **Client Dashboard**

1.3.2 Client Dashboard Operations

20. Client can perform the following tasks:

- View Documents
- Download Documents
- Select documents by category
- Select documents by year

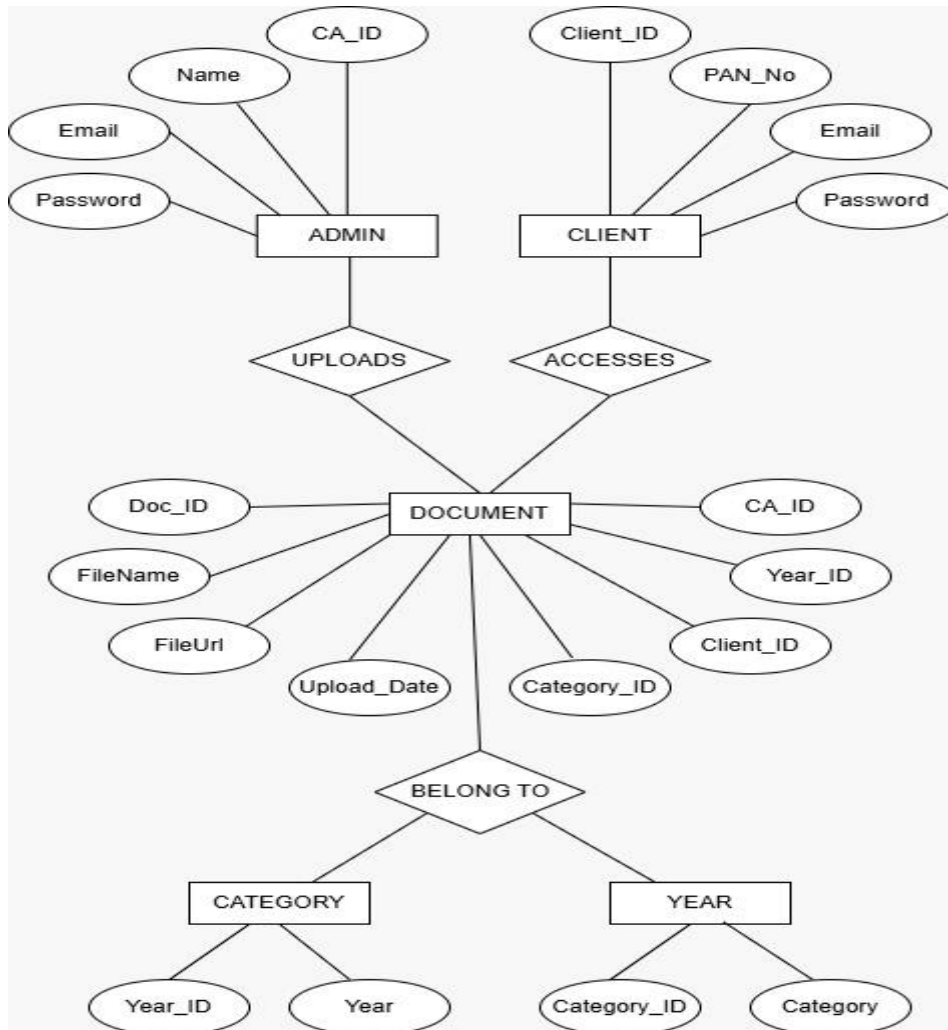
21. After completing work, Client clicks **Logout**.

22. System ends.

1.3.3 End of Flowchart

23. **Stop / End**

5.3 Entity Relationship Diagram



This ER diagram represents a CA Desk website used to manage client documents securely between Admin (CA) and Clients.

1. ADMIN Entity

The ADMIN entity represents the Chartered Accountant who manages clients and uploads documents.

Attributes:

CA_ID (Primary Key): Unique identifier of the admin

Email: Admin email address

Password: Login password

Relationship:

ADMIN uploads documents

→ One admin can upload multiple documents (1 : M relationship with DOCUMENT)

2. CLIENT Entity

The CLIENT entity represents clients registered under the CA.

Attributes:

Client_ID (Primary Key): Unique client identifier

Email: Client email address

Password: Client login password

Relationship:

CLIENT accesses documents

→ One client can access multiple documents

3. DOCUMENT Entity

The DOCUMENT entity stores details of uploaded files.

Attributes:

Doc_ID (Primary Key): Unique document identifier

FileName: Name of the document

FileUrl: Location/path of the document

Upload_Date: Date of upload

CA_ID (Foreign Key): Admin who uploaded the document

Client_ID (Foreign Key): Client to whom the document belongs

Category_ID (Foreign Key): Category of the document Year_ID

(Foreign Key): Financial year of the document Relationships:

Document is uploaded by ADMIN

Document is accessed by CLIENT

Document belongs to CATEGORY and YEAR

4. CATEGORY Entity

The CATEGORY entity classifies documents based on type.

Attributes:

Category_ID (Primary Key)

Category (e.g., GST, IT Return, Audit)

Relationship:

One category can have many documents

Each document belongs to one category

5. YEAR Entity

The YEAR entity represents the financial year of documents.

Attributes:

Year_ID (Primary Key)

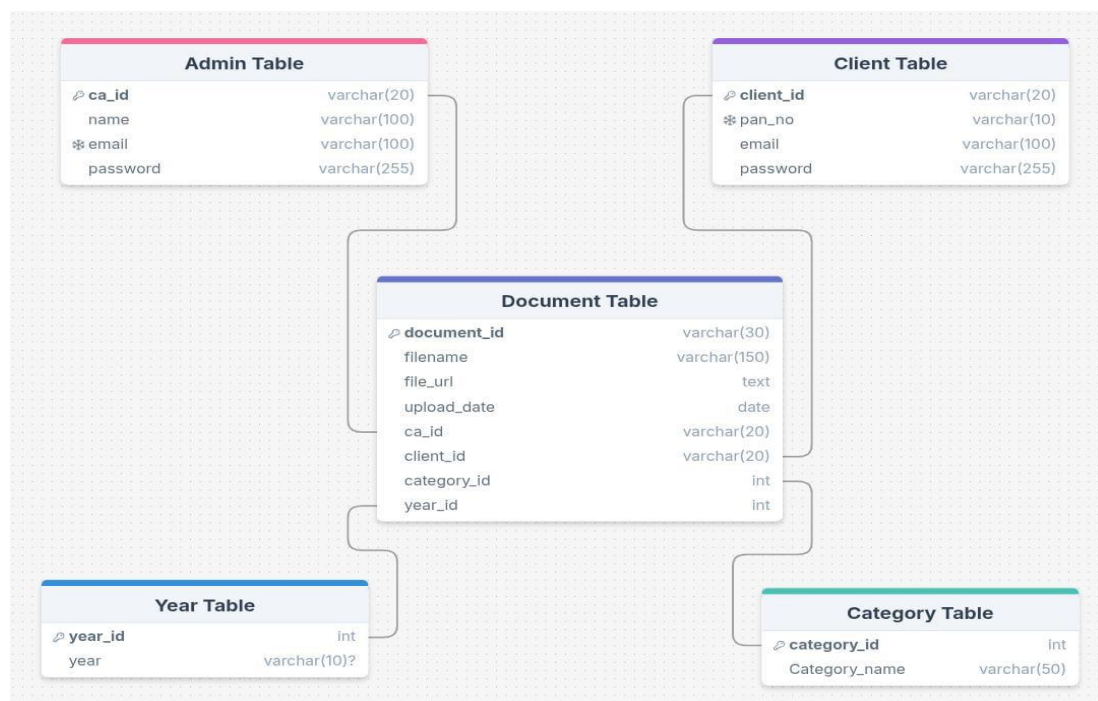
Year (e.g., 2022–23, 2023–24)

Relationship:

One year can have many documents

Each document belongs to one year

5.4 Data Schema



Description of the Diagram

The given diagram represents the Database Schema Diagram of the “CA Desk.”

It shows the structure of the database, the tables used in the system, their attributes, and the relationships between them.

The diagram consists of five tables:

1. Admin
2. Client
3. Document
4. Year
5. Category

Each table has a primary key to uniquely identify records, and foreign keys are used to connect related tables. The diagram mainly represents one-to-many relationships between the tables.

Description of Each Table

1. Admin Table

Description:

This table stores details of the Chartered Accountant or administrator who manages the system.

Attributes:

ca_id, name, email, password

Purpose:

The purpose of the Admin table is to maintain login credentials and personal details of the CA so that only authorized admins can manage clients and documents.

2. Client Table

Description:

This table stores information about clients registered under a CA.

Attributes:

client_id, pan_no, email, password

Purpose:

The purpose of the Client table is to store client login details and identity information so that clients can securely access their own documents.

3. Document Table

Description:

This is the main table of the system that stores all uploaded documents.

Attributes:

document_id, filename, file_url, upload_date, ca_id, client_id, category_id, year_id

Purpose:

The purpose of the Document table is to keep records of all client documents and link them with the respective admin, client, category, and year.

4. Year Table

Description:

This table stores financial year details.

Attributes:

year_id, year

Purpose:

The purpose of the Year table is to organize and categorize documents according to different financial years.

5. Category Table

Description:

This table stores different document categories.

Attributes:

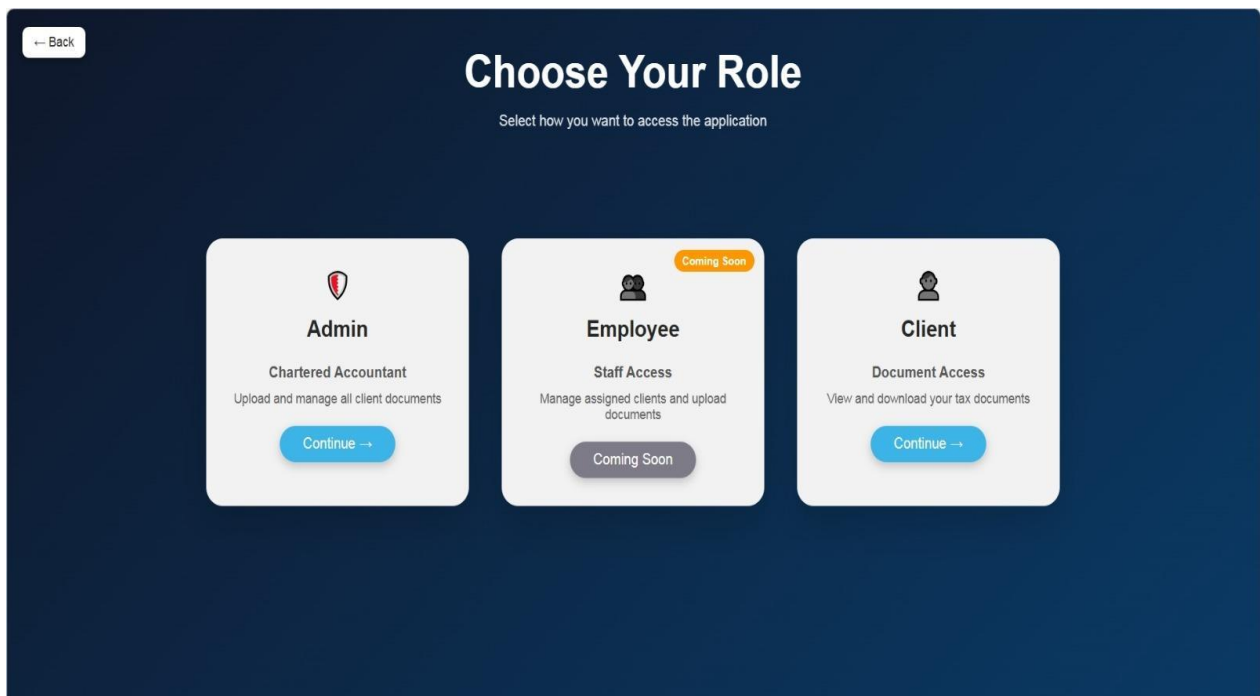
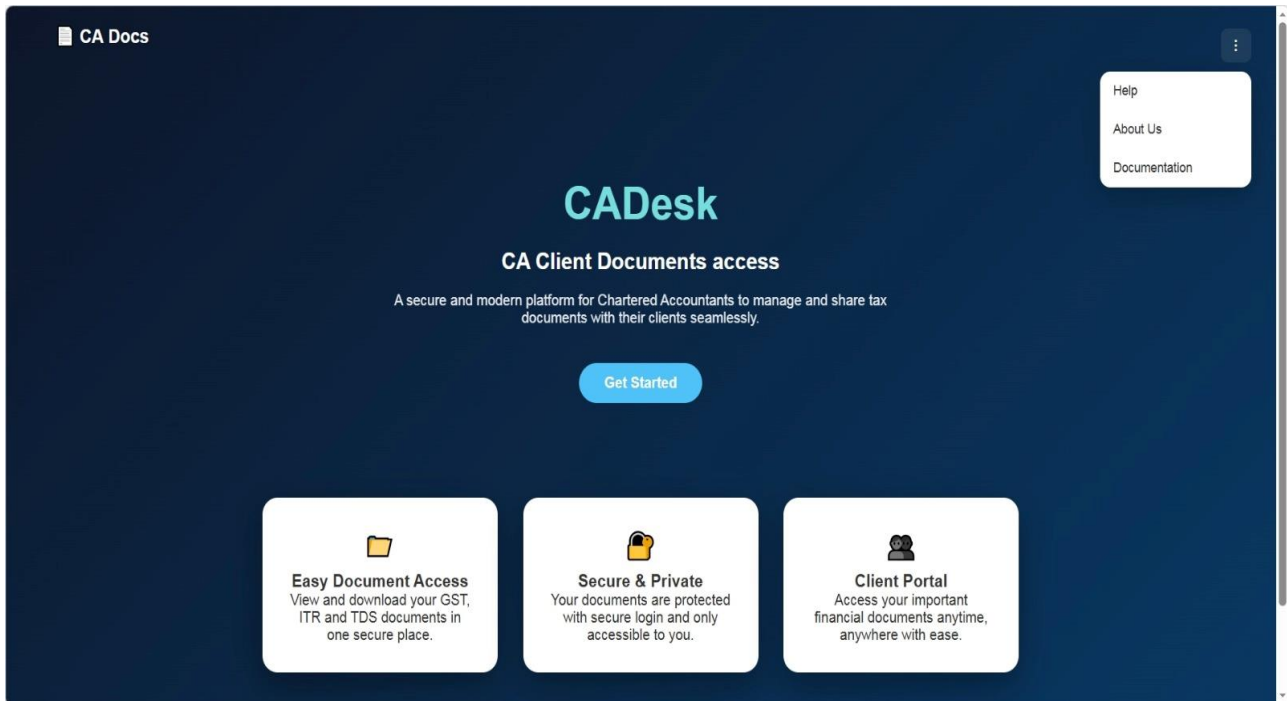
category_id, category_name

Purpose:

The purpose of the Category table is to classify documents based on their type such as GST, ITR, TDS, etc.

These relationships are implemented using foreign keys in the Document table.

5.5 Screenshot of UI



[← Back](#)

Admin Login

Access your admin dashboard

Login

Welcome back,

CA Admin



+ New Client

+ New Employee

3

Total Clients

5

Total Documents

Quick Actions

Upload Documents

View Clients

Document Categories

GST



GST Returns & Filings

2 >

ITR



Income Tax Returns

1 >

Add New Client

Create Client

Upload Document

Select Client

Service

Document Title

Year

Google Drive Link

Upload Document

[← Back](#)

Client Login

Access your client dashboard

Login

My Documents

[Dashboard](#)

[Profile](#)

[Logout](#)

Welcome, Rahul Sharma

Access and download your tax documents below.

GST Documents

1

[View All](#)

ITR Documents

1

[View All](#)

TDS Documents

2

[View All](#)

Recent Documents

TDS Statement 2026

[View](#)

ITR Filing

[View](#)

TDS Statement 2024

[View](#)

GST Return 2024

[View](#)

Select Documents

Last 1 Year

Last 2 Years

Both Years

Your Documents

TDS Statement 2024
Year : 2024

View & Download

TDS Statement 2026
Year : 2026

View & Download

6. System Coding

6.1 Code

1) About Page (about.html)

This page provides information about the CA Desk system and its purpose.

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>About Us</title>
```

```
<link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
<div class="container">
```

```
<h1>About CA Desk</h1>
```

```
<p>
```

CA Desk is a web-based system developed to help Chartered Accountants and their clients manage important documents securely.

```
</p>
```

```
<p>
```

Clients can access and download documents anytime without visiting the office.

```
</p>
```

```
</div>
```

```
</body>
```

```
</html>
```

2) Help Page (help.html)

This page provides basic guidance to users on how to use the system.

```
<!DOCTYPE html>.
```

```
<html>
```

```
<head>
```

```
<title>Help</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<div class="help-section">
<h1>Help Center</h1>
<ul>
<li>Login using your registered email.</li>
<li>Select your dashboard.</li>
<li>Download your documents securely.</li>
</ul>
</div>
</body>
</html>
```

3) Documentation Page (documentation.html)

This page explains system documentation and working of the project.

```
<!DOCTYPE html>
<html>
<head>
<title>Documentation</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<div class="docs">
<h1>System Documentation</h1>
<p>This system stores client documents securely using Firebase database.</p>
</div>
</body>
</html>
```

4) Navigation Menu Script

This code controls the navigation menu display.

```
javascript
const menuButton = document.getElementById("menuBtn");
const menu = document.getElementById("dropdownMenu");

menuButton.addEventListener("click", function() {
  if(menu.style.display === "block"){
    menu.style.display = "none";
  }
  else{
    menu.style.display = "block";
  }
});
```

5) Create Document Card JS

This function dynamically creates document cards on the dashboard.

```
javascript
function createDocumentCard(title,link){
  const card = document.createElement("div");
  card.className = "document-card";
  card.innerHTML = `
<h3>${title}</h3>

<a href="${link}" target="_blank">Download</a>;
documentsContainer.appendChild(card);
}
```


6) HTML Code (Login Page Structure)

This HTML code creates the login page interface where users enter their email and password to access the CA Desk system.

```
<!DOCTYPE html>
<html>
<head>
<title>CA Desk Login</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<div class="login-container">
<h2>Login</h2>
<input type="email" id="email" placeholder="Enter Email">
<input type="password" id="password" placeholder="Enter Password">
<button id="loginBtn">Login</button>
</div>
<script type="module" src="login.js"></script>
</body>
</html>
```

7) CSS Code (Website Styling)

This CSS code styles the login page and provides a clean and professional interface for the CA Desk website.

```
body{
font-family: Arial;
background-color:#f5f5f5;
}
.login-container{
width:300px;
margin:auto;
```

```

margin-top:120px;
padding:20px;
background:white;
border-radius:8px;
box-shadow:0 0 10px rgba(0,0,0,0.1);
}
input{
width:100%;
padding:10px;
margin:10px 0;
}
button{
width:100%;
padding:10px;
background:#1a73e8;
color:white;
border:none;
border-radius:5px;
}

```

8) Login Authentication System (login.js)

This code authenticates users using Firebase login and redirects them to the correct dashboard based on their role (Admin, Client, or Employee).

```

import { auth, db } from "./firebase.js";
import { signInWithEmailAndPassword }
from "https://www.gstatic.com/firebasejs/10.12.0/firebase-auth.js";
import { doc, getDoc }
from "https://www.gstatic.com/firebasejs/10.12.0/firebase-firestore.js";
document.getElementById("loginBtn").addEventListener("click", async function () {
const email = document.getElementById("email").value.trim();

```

```

const password = document.getElementById("password").value.trim();
const user redential = await signInWithEmailAndPassword(auth, email, password);
const user = userCredential.user;
const userDoc = await getDoc(doc(db, "users", user.uid));
const role = userDoc.data().role;
if (role === "admin") {
  window.location.href = "admin-dashboard.html";
}
else if (role === "client") {
  window.location.href = "client-dashboard.html";
}
else if (role === "employee") {
  window.location.href = "employee-dashboard.html";
}
});

```

9) Firebase Configuration ([firebase.js](#))

This code connects the CA Desk website with Firebase services such as Authentication and Firestore Database.

// Import Firebase

```

import { initializeApp } from "https://www.gstatic.com/firebasejs/10.12.0/firebase-app.js";
import { getAuth } from "https://www.gstatic.com/firebasejs/10.12.0/firebase-auth.js";
import { getFirestore } from "https://www.gstatic.com/firebasejs/10.12.0/firebase-firestore.js";

```

// Firebase Config

```

const firebaseConfig = {
  apiKey: "AIzaSyBJFUymB70otlMgXX0uF2faH3kUxKnT6ms",
  authDomain: "cadesk-51b66.firebaseio.com",
  projectId: "cadesk-51b66",
  storageBucket: "cadesk-51b66.firebaseio.com",

```

```

messagingSenderId: "123347321836",
appId: "1:123347321836:web:dff6571c011bea213cf6c9"
};

const app = initializeApp(firebaseConfig);
export const auth = getAuth(app);
export const db = getFirestore(app);

```

10) Upload Document Module ([upload-document.js](#))

This function loads all clients from the Firestore database and displays them in a dropdown list when uploading documents.

```

import { db } from "../firebase.js";
import {
  collection,
  getDocs,
  addDoc
} from "https://www.gstatic.com/firebasejs/10.12.0/firebase-firestore.js";
const clientSelect = document.getElementById("clientSelect");
const uploadBtn = document.getElementById("uploadBtn");

// Load clients
async function loadClients(){
  const snapshot = await getDocs(collection(db,"clients"));
  snapshot.forEach((doc)=>{
    const data = doc.data();
    const option = document.createElement("option");
    option.value = doc.id;
    option.textContent = data.name;
    clientSelect.appendChild(option);
  });
}

loadClients();

```

11) View Clients Module ([view-clients.js](#))

This code fetches all client records from the database and displays them on the View Clients page.

```
import { db } from "../firebase.js";
import {
  collection,
  getDocs
} from "https://www.gstatic.com/firebasejs/10.12.0/firebase-firestore.js";
const container = document.getElementById("clientContainer");

async function loadClients(){
  const snapshot = await getDocs(collection(db,"clients"));
  for(const clientDoc of snapshot.docs){
    const data = clientDoc.data();
    const div = document.createElement("div");
    div.innerHTML = `<h3>${data.name}</h3>`;
    container.appendChild(div);
  }
  loadClients();
}
```

6.2 Validation Code

```
document.getElementById("loginBtn").addEventListener("click", async function () {
  const email = document.getElementById("email").value.trim();
  const password = document.getElementById("password").value.trim();
  const errorBox = document.getElementById("errorMessage");

  errorBox.style.display = "none";
  errorBox.textContent = "";
```

```
// Validation
if (!email || !password) {
  errorBox.style.display = "block";
  errorBox.textContent = "Please enter email and password";
  return;
}

try {
  const userCredential = await signInWithEmailAndPassword(auth, email, password);
  const user = userCredential.user;

  const userDoc = await getDoc(doc(db, "users", user.uid));

  if (!userDoc.exists()) {
    errorBox.style.display = "block";
    errorBox.textContent = "User data not found in database";
    return;
  }
} catch (error) {
  errorBox.style.display = "block";
  errorBox.textContent = "Invalid email or password";
}
});
```

7. System Testing

7.1 Definition of Quality Assurance (QA):

Quality Assurance is the process of checking and improving the methods used to develop a product or software to make sure it meets the required quality standards and works correctly without defects.

System Testing :

System Testing is a type of software testing where the entire system is tested as a whole.

It is done after integration testing.

The main purpose is to check whether the system meets the specified requirements.

It ensures that all components work together correctly without errors.

1. Black Box Testing

Black Box Testing is a software testing technique in which the tester tests the functionality of the software without knowing the internal code, structure, or implementation of the program. The tester only checks the input given to the system and the output produced by the system.

The main aim of black box testing is to verify whether the system works according to the specified requirements. In this method, the tester behaves like a normal user of the system and checks whether the software performs the expected functions.

Example:

If a user enters a correct username and password on a login page and the system opens the dashboard, the system is working correctly. The tester does not need to know how the code inside the login program works.

Types of Black Box Testing

1. Equivalence Partitioning – Input data is divided into different groups (valid and invalid), and one value from each group is tested.
2. Boundary Value Analysis – Tests the values at the minimum and maximum limits of input.
3. Decision Table Testing – Tests different combinations of conditions and actions.
4. State Transition Testing – Checks how the system behaves when it changes from one state to another.
5. Error Guessing – Tester predicts possible errors based on experience.

2. White Box Testing

White Box Testing is a software testing technique in which the tester has complete knowledge of the internal code, program structure, and logic of the system. The tester checks how the program works internally and ensures that all the code statements, conditions, and paths are executed correctly.

The main aim of white box testing is to verify the internal working of the program and improve code quality. This testing is usually performed by developers or testers who have programming knowledge.

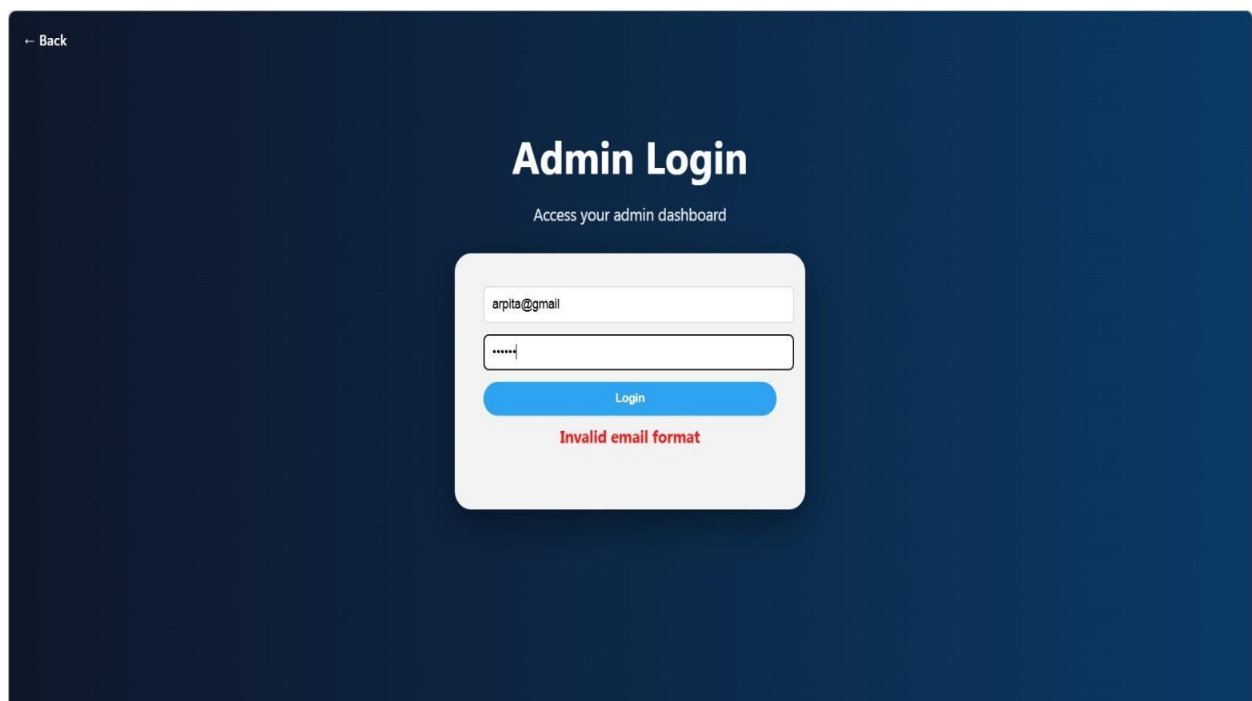
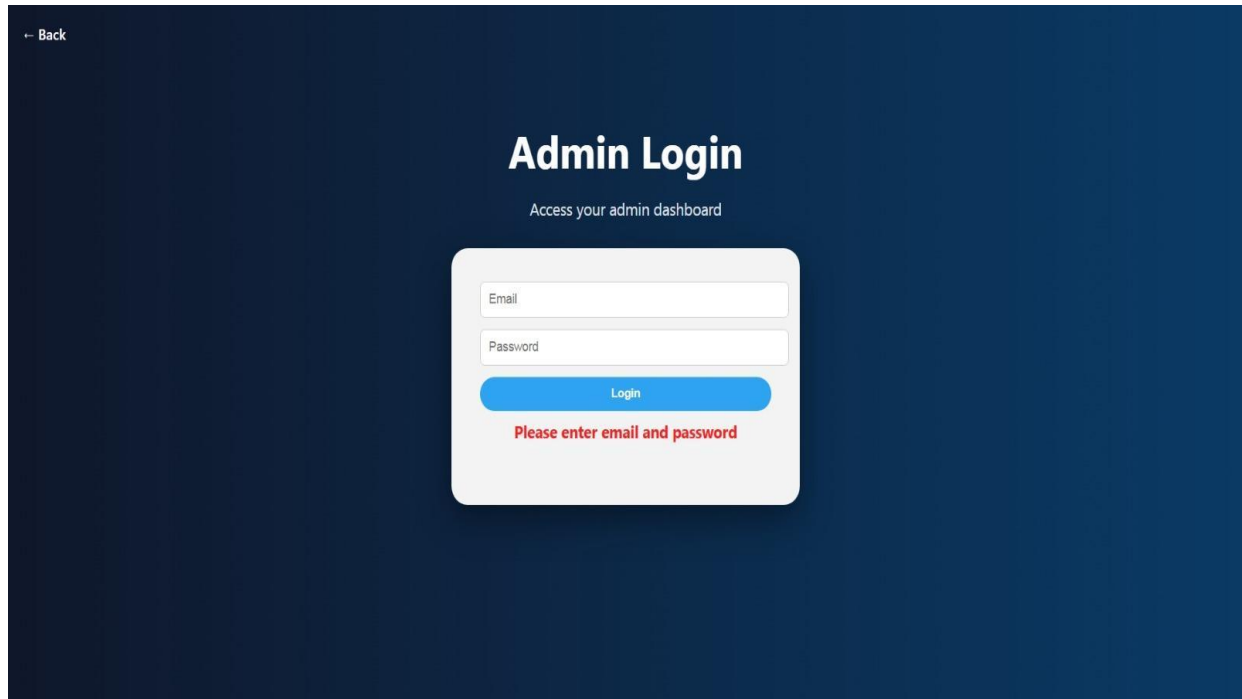
Example:

A programmer checks whether all the loops, conditions (if-else), and functions in the code are executed properly.

Types of White Box Testing

1. Statement Coverage – Ensures that every statement in the program is executed at least once.
2. Branch Coverage – Ensures that all possible branches (true and false conditions) in the code are tested.
3. Path Coverage – Tests all possible execution paths in the program.
4. Condition Coverage – Tests each logical condition in the code.

7.2 Screenshot of Testing



Add New Client

Name required

Email required

Password required

Create Client

Add New Client

Enter valid email

Password must be at least 6 characters

Create Client

← Back

Client Login

Access your client dashboard

admin@gmail.com

Login

Invalid email or password

127.0.0.1:5501 says

Are you sure you want to logout?

OK

Cancel



Client Profile



Name
Rahul Sharma



Email
rahul@gmail.com

Logout

8. Implementation

8.1 Installation steps

Before installing and deploying the CA Client Documents Access Website, some preparation steps are performed to ensure the system works properly.

Step 1: System Requirement Check

The system is checked to ensure it has a compatible web browser such as Chrome, Edge, or Firefox. It should also have sufficient storage, RAM, and a stable internet connection.

Step 2: Development Environment Setup

Required software such as Visual Studio Code (VS Code) and programming environments like Java and Python are installed and configured for running the website.

Step 3: Server Preparation

A local or hosting server is prepared so that the website can run and users can access it through a web browser.

Step 4: Database Configuration

The database is prepared to store important data such as client login details and uploaded documents securely.

Step 5: Website Files Preparation

All project files including HTML, CSS, JavaScript, Java, and Python are organized and checked to ensure there are no errors.

Step 6: User Account Setup

Admin and client login accounts are created so that authorized users can access the system securely.

Step 7: Network Check

The internet and network connection are checked to ensure smooth communication between the website and the server.

8.2 User Guide

The User Guide provides instructions on how users can access and use the CA Client Documents Access Website. It helps clients and administrators understand how to perform different tasks in the system.

Step 1: Open the Website

The user first opens a web browser such as Chrome or Edge and enters the website URL to access the system.

Step 2: Login to the System

The user enters their username and password on the login page. After entering the correct credentials, the user can access their dashboard.

Step 3: Dashboard Access

After login, the dashboard page appears. From the dashboard, users can view different options such as documents, profile details, and notifications.

Step 4: View Documents

Clients can click on the Documents section to view the files uploaded by the CA. The documents are organized for easy access.

Step 5: Download Documents

Users can select a document and click the Download button to save the file to their device.

Step 6: Logout

After completing their work, the user can click the Logout button to safely exit the system.

9.Conclusion

The CA Client Documents Access Website was developed to simplify the process of managing and accessing important financial documents between Chartered Accountants and their clients. In the traditional method, clients usually need to visit the CA office or communicate through email or messaging applications to receive documents, which can be time-consuming and sometimes unorganized. This project provides a digital solution to that problem.

We chose this project because many CA offices handle a large number of documents such as GST files, tax returns, financial statements, and other important records. Managing these documents manually can lead to delays, confusion, or loss of files. By creating this website, documents can be stored in one place and accessed easily whenever required.

The system allows the admin (CA or staff) to upload and manage documents for different clients, while the clients can securely log in to view or download their documents. This improves communication between the CA and clients and makes document sharing faster and more organized.

Overall, this project helps in saving time, improving document security, reducing paperwork, and making document management more efficient. It also demonstrates how web technology can be used to solve real-world problems in professional services like accounting and taxation.

10.Future Enhancement

The CA Client Documents Access Website can be further improved in the future by adding more advanced features to make the system more efficient, secure, and user-friendly.

1. Mobile Application

In the future, a mobile application (Android and iOS) can be developed for the system. This will allow clients to access their documents easily through their smartphones anytime and anywhere, making the service more convenient.

2. Automated Notifications

The system can include automatic email or SMS notifications. Whenever a new document is uploaded by the admin, the client will receive an instant alert. This will help clients stay updated without checking the website frequently.

3. Enhanced Security

Security can be improved by implementing OTP-based login and Multi-Factor Authentication (MFA). This will provide an additional layer of protection and ensure that only authorized users can access important financial documents.

4. Employee Dashboard

In the future, an employee dashboard can be added where CA office staff can manage tasks, track client work, and organize documents more efficiently. This will help in improving workflow and productivity within the CA office.

11. Reference of bibliography

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